

Density + Rearranging Equations

Quick reference sheet · Physics & Maths · keep handy for revision

1. What density measures

Density is how much mass is packed into a given amount of space. Two objects can be the same size but have very different densities if one has more mass squeezed into it.

density = mass ÷ volume or $\rho = m / V$

- ρ (rho) — density, in kg/m^3 or g/cm^3
- m — mass, in kg or g
- V — volume, in m^3 or cm^3

2. Rearranging the equation

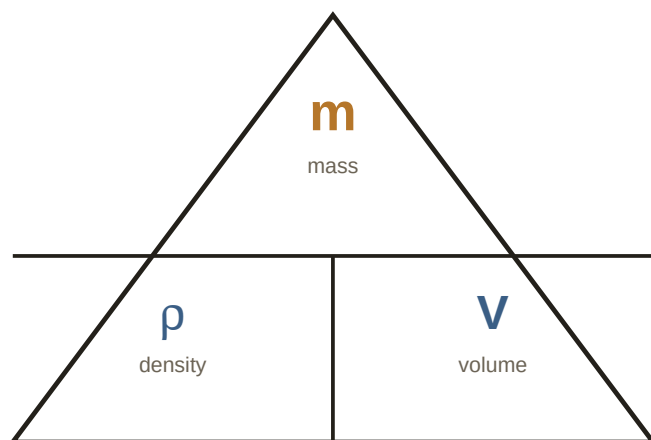
Whatever you do to one side of an equation, you must do to the other. Starting from $\rho = m / V$:

To find **mass**: multiply both sides by $V \rightarrow m = \rho V$

To find **volume**: multiply both sides by V , then divide both sides by $\rho \rightarrow V = m / \rho$

This method works on every equation in the course — worth doing properly rather than memorising three separate formulas.

3. Formula triangle (shortcut)



Cover the letter you want to find. The remaining two letters show the operation: top-over-bottom means divide, side-by-side on the bottom means multiply.

Cover **m** → see ρ next to $V \rightarrow m = \rho V$. Cover ρ → see m over $V \rightarrow \rho = m / V$. Cover **V** → see m over $\rho \rightarrow V = m / \rho$.

4. Worked example

Given	Find	Working	Answer
$m = 20 \text{ kg}, V = 4 \text{ m}^3$	ρ	$\rho = m / V = 20 / 4$	5 kg/m^3
$\rho = 8 \text{ g/cm}^3, V = 5 \text{ cm}^3$	m	$m = \rho V = 8 \times 5$	40 g
$\rho = 2 \text{ g/cm}^3, m = 16 \text{ g}$	V	$V = m / \rho = 16 / 2$	8 cm^3

Tip: always write the rearranged equation as its own line before substituting numbers — that step earns a method mark even if the final arithmetic goes wrong.